ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

# Comparative fMRI Analysis of Hand and Foot Movement Representations

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Mini Project 1 Report

NX421 - Neural Signals and Signal Processing

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# 1 Part I: Preprocessing and GLM

## 1.1 Structural Preprocessing

### 1.1.1 Skull-stripping

Skull-stripping involves isolating brain from non-brain tissue like the skull and scalp (Fig. 1.1) . We achieved this done by creating a binary mask using FSL's Brain Extraction Tool (BET) in the T1 weighted anatomical image. This gives us an anatomical image of only brain tissue. (Fig. 1.2)

### 1.1.2 Segmentation

Segmentation separates brain tissues into different tissue types: gray matter, white matter, and cerebrospinal fluid, for more precise anatomical boundaries. We used FMRIB's Automated Segmentation Tool (FAST) in FSL to classify voxels into different tissue types based on their intensity distribution. In the color overlay on the anatomical image (Fig. 1.3), blue regions correspond to the cerebrospinal fluid, red regions to grey matter and green regions to white matter.

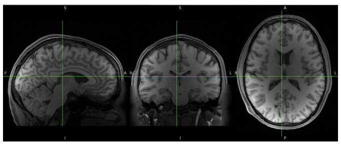


**Figure 1.1**
Original T1-weighted image.

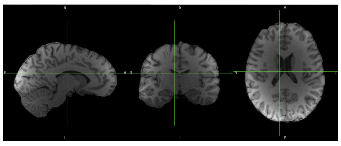


**Figure 1.2**
Brain-extracted image obtained.

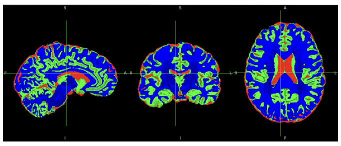


**Figure 1.3**
Tissue segmentation by FAST.

## 1.2 Functional Preprocessing

### 1.2.1 Concatenation of Runs

We used a brain mask to select for voxels with mean absolute intensity > $10^{-6}$ to exclude background zeros and only retain all real voxels for both runs of subject 101410. We then normalized the data from both runs using the global variance, which rescaled the variance to 1. We then concatenated the two runs to get a single time series dataset with 568 timepoints.

### 1.2.2 Motion Correction

Motion correction is essential because even small head movements during scanning can cause significant artifacts that degrade data quality and bias results, especially in experiments like ours where motion may correlate with the experimental task. We performed motion correction using McFLIRT. By default, McFLIRT chooses the middle volume of the timeseries data as the reference volume but since our data is a concatenation of the 2 runs, the middle volume would be the first volume of the second run and is therefore likely to be noisy. So we attempted motion correction with the middle volumes of both the runs and with the average volume of the data as the reference. They gave similar results so arbitrarily chose the average volume as the reference. We found 26 highly irregular volumes (Fig. 1.4) as defined by The threshold was computed as Threshold $= 2 \times Q_{0.75} + 1.5 \times (Q_{0.75} - Q_{0.25})$.

However, we decided to not drop them as that would result in our fMRI data not being correctly aligned to the event markers in the experiment. We handled these volumes by later including them as regressors in the GLM.

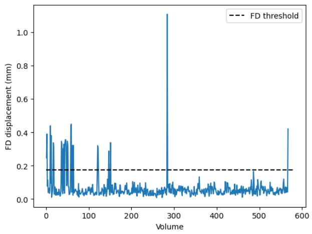



**FIGURE 1.4**
Frame-wise displacement in consecutive volumes

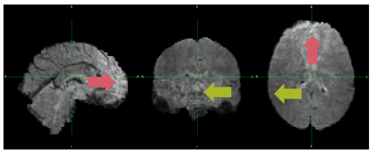


**FIGURE 1.5**
Regions where motion correction had the greatest effect.

Visualising motion correction in FSLeyes and through motion parameters revealed that the strongest motion correction effect was found along the negative $x$-axis (green arrows in Fig. 1.5) and that volumes identified as outliers were primarily associated with translations along the $y$-axis (see red arrows in Fig. 1.5).

### 1.2.3 Gaussian Smoothing

Spatial smoothing involves averaging the signal of each voxel with its neighbors using a Gaussian kernel to boost the signal to noise ratio (SNR). It also increases spatial correlation to match the size of expected brain activation patterns over multiple voxels. The kernel size must be chosen to balance the tradeoff between the loss of spatial precision and the boost in SNR and is usually set by a FWHM twice the voxel size. Since our voxels were 2mm X 2mm X 2mm, we used an FWHM of 4mm. (Fig. 1.6)(Fig. 1.7)

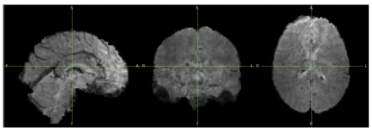


**FIGURE 1.6**
Before Gaussian smoothing.

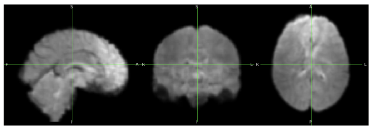


**FIGURE 1.7**
After Gaussian smoothing.

## 1.3 GLM Analysis

### 1.3.1 Experimental Design Matrix

We used the preprocessed event files and the 'FirstLevelModel' function to generate a suitable design matrix where each column represents a predictor of the BOLD signal.
Task regressors (HRF-convolved): `cue`, `left_hand`, `right_hand`, `left_foot`, `right_foot`, `tongue`, `rest`
Nuisance regressors: Polynomial drifts (`drift_1`, `drift_2`, `drift_3`), Constant term
Motion outlier regressors: one binary column per high-motion scan detected in motion correction. This ensures that these volumes are excluded from GLM fitting and do not bias estimates. Fig. 1.9)

### 1.3.2 First-level GLM Analysis

We ran a first-level GLM as it is used for within-subject statistical modeling of brain activity at each voxel over time, since our dataset covers only one subject.

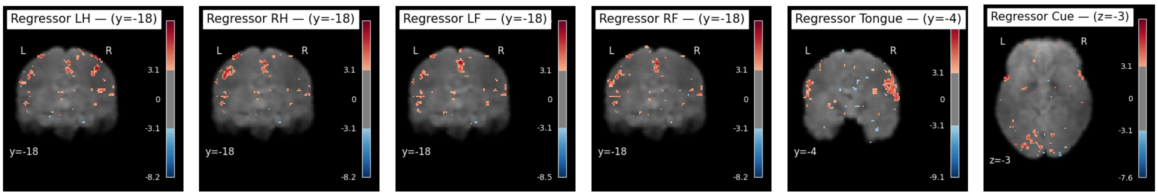
Statistical maps of each task-related regressor represent voxel-wise estimates of significantly increased/decreased $|Z| \geq 3.09$ brain activation associated with that experimental condition in comparison to baseline from the fitted GLM. The regions of maximal activation for each task-related regressor are plotted.

**Left Hand/Right Hand:** MNI $y = -18$ (coronal plane) Activations found in precentral gyrus (premotor and primary motor cortex M1) and postcentral gyrus (somatosensory cortex S1, $y = -22$) which exhibit somatomotor homuncular organization and contralateral motor control.

**Left Foot/Right Foot:** MNI $y = -18$ (coronal plane) Activation found in the precentral gyrus corresponding to contralateral motor cortex engagement and more medial activity in the M1 and S1 cortices ($y = -22$ - $-24$) than hands, which corresponds to the leg and foot areas instead of the more lateral hand areas in the somatotopic organization of the sensorimotor cortex.

**Tongue:** MNI $y = -4$ (coronal plane) Activation on the lateral surface of the precentral gyrus, corresponding to inferior portion M1 and aligning with the lateral orofacial and tongue representations in the motor homunculus.

**Cue (MNI $z = -3$, axial plane)** Activations more diffuse and of lower intensity, observed primarily in occipital and temporal regions, consistent with visual processing during cue presentation.




**(A)** LH **(B)** RH **(C)** LF **(D)** RF **(E)** Tongue **(F)** Cue

**FIGURE 1.8**
Task-related regressor activation maps (single row).

### 1.3.3 CONTRAST DESIGN AND ACTIVATION MAPS

We designed 2 sets of contrast vectors to generate activation maps for hand against foot motion.

1) A simple average hand vs foot contrast vector: $0.5\,[(\mathrm{RH}+\mathrm{LH})-(\mathrm{RF}+\mathrm{LF})]$.

2) Set of vectors to include additional information from tongue movement to hopefully capture artifacts for processing the cue, making a decision, and movement intention:

(I) Avg Hands $>$ Avg Foot + Tongue: $(\mathrm{RH}+\mathrm{LH})-[0.5\,(\mathrm{LF}+\mathrm{RF})+\mathrm{T}]$

(II) Avg Foot $>$ Avg Hands + Tongue: $(\mathrm{RF}+\mathrm{LF})-[0.5\,(\mathrm{LH}+\mathrm{RH})+\mathrm{T}]$

and subtracting $I - II$ to obtain Avg Hand $>$ Avg Foot.

However, we chose to exclude second vector from the final analysis as the results were very similar to the first simple contrast, and because the activation maps obtained for $I$ and $II$ were not easily interpretable.

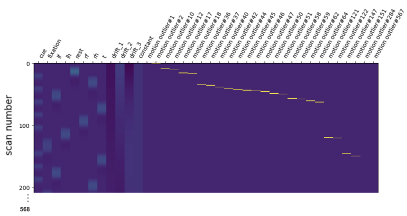



**FIGURE 1.9**
Design matrix visualization

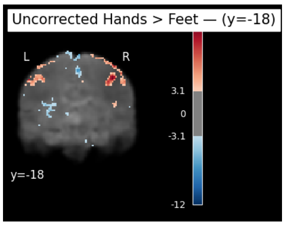



**FIGURE 1.10**
**Uncorrected:** voxelwise $p < 0.001$ ($Z \geq 3.09$).

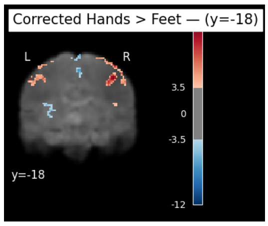



**FIGURE 1.11**
**Corrected:** FDR $q < 0.05$, cluster $> 10$ voxels.

Our results match the expected organization of hand and foot representation in the sensorimotor cortex (postcentral gyrus): more lateral activation for hands and medial activation for feet.

### 1.3.4 ATLAS OVERLAY AND REGIONAL INTERPRETATION

We overlaid our contrast maps with the AAL atlas parcellation using `NiLearn` to identify the anatomical regions corresponding to maximal activation. We plotted both *Hands* $>$ *Feet* and *Feet* $>$ *Hands* contrasts as a sanity check.

**Hands $>$ Feet contrast:** Peak activations were observed in right M1 at $(44, -18, 50)$ and S1 at $(56, -18, 48)$, with the largest cluster in right M1 ($Z = 12.30$; $2032\ \mathrm{mm}^3$). Left and right M1 are involved in the initiation and control of precise, voluntary movements. Additional activity in the superior parietal lobule and cerebellum likely reflects contributions to spatial orientation and motor coordination, respectively. Fig. 1.12, 1.13, 1.14)

**Feet $>$ Hands contrast:** Strongest activations observed in medial motor regions, including the supplementary motor area (SMA) at $(-6, -10, 74)$, paracentral lobule at $(-6, -38, 72)$, and mid-cingulate cortex at $(-10, 2, 44)$. These regions are associated with lower limb movement, motor preparation, and postural coordination. The largest cluster was in the left SMA ($Z = 11.35$; $4944\ \mathrm{mm}^3$). Both left and right SMAs, which lie medially and anterior to M1, are known to be involved in higher-order motor planning. Fig. 1.15, 1.16, 1.17)

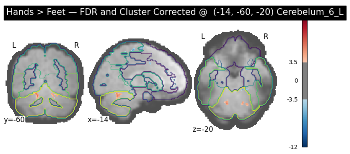



**FIGURE 1.12**
Hands $>$ Feet (Cerebellum L.)

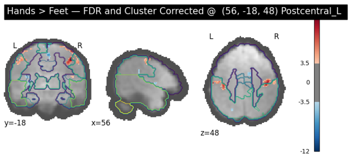



**FIGURE 1.13**
Hands $>$ Feet (Postcentral Gyrus L.)

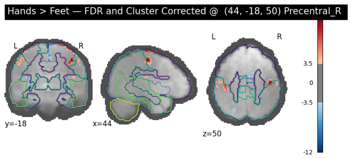



**FIGURE 1.14**
Hands $>$ Feet (Precentral Gyrus R.)

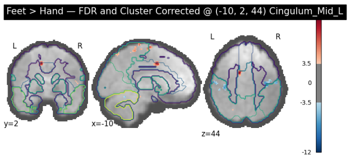



**FIGURE 1.15**
Feet $>$ Hands (Cingulum Mid L.)

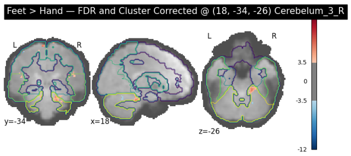



**FIGURE 1.16**
Feet $>$ Hands (Cerebellum R.)

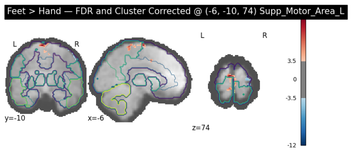



**FIGURE 1.17**
Feet $>$ Hands (Supplementary Motor Area L.)

## 1.4 Theoretical Questions I

### 1.4.1 Answer 1

Second-level analysis assesses inter-subject variability. We have only a single subject in the dataset provided, so we cannot use it to perform a second level analysis.

### 1.4.2 Answer 2

Yes, we could perform a second-level analysis if provided an extended dataset with 10 subjects and their age information, for the same task. We could also use this to address additional questions, like the effect of age on the differential neuronal activation in hand versus foot movement.
To achieve this, we could use the simple Avg Hand > Avg Foot contrast vector for the first level analysis.
For the second-level analysis, we could z-score age and create a second-level analysis matrix with one column for the intercept (all 1's, testing the average effect across subjects) and another for age, z-scored across subjects as a covariate (Note: If the sample were larger, we could perform a one-way ANOVA.)
We can now use the contrast: c $= [0_{\text{intercept}}, 1_{\text{age}}]$ to also investigate the effect of age on motor control.

# 2 Part II: PCA

## 2.1 PCA Analysis

For a data-driven analysis of our functional data, we applied PCA on the given fMRI runs, considering each volume as a sample, using the `scikit-learn` library in Python. We used incremental PCA to perform this in batches to prevent memory errors due to the large size of the data.

We chose to retain the first $15$ PCA components as they explained over $85\%$ of the variance. We did not go up to $90\%$ of the variance, as that would involve around $50$ components.

We used the Pearson correlation coefficient as our similarity measure, as it measures the linear correlation between the component vectors. It ranges from $[-1, 1]$, with $1$ being perfectly positively correlated, $-1$ being perfectly negatively correlated, and $0$ implying no correlation. We obtained a $5 \times 5$ similarity with all diagonal elements $1$, as each component is perfectly correlated with itself. The off-diagonal elements have very low values, suggesting that the components are orthogonal or unrelated and each component captures unique information.

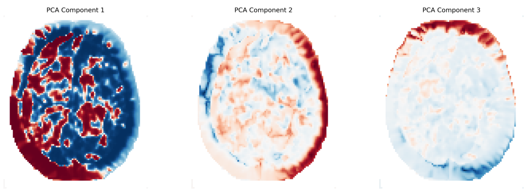



**FIGURE 2.1**
Axial view of the first three PCA components.

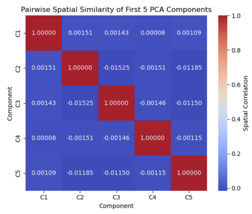



**FIGURE 2.2**
Similarity matrix of the first five PCA components.

## 2.2 THEORETICAL QUESTIONS II

### 2.2.1 ANSWER 1

Given the motor task performed and visual cues presented, we expected activity in the visual and somatomotor networks, with some activity in the default mode network. The strongest principal component appears to be representative of contralateral activation of the brain as both hemispheres are strongly oppositely activated. This may arise from the directional nature of the task instead of a functional network. The third component shows activity in the occipital lobe, suggesting activation of the visual network (Yeo et al. 2011). Additional activity in the prefrontal cortex leads us to speculate that this component may correspond to a subject processing a visual cue and deciding which body part to move. Other components show diffuse activation and are hard to interpret.

### 2.2.2 ANSWER 2

In GLM, we observed significant activations in the primary motor cortex and somatosensory cortex. These regions are known to exhibit contralateral control which may correspond to the activity captured by the first PCA component which shows left–right hemispheric differentiation. However, no PCA component clearly isolates the somatomotor network. The statistical map for the cue regressor in GLM shows activity in the occipital lobe like in PC3.

The GLM results clearly identify expected activations in precentral and postcentral gyri, as well as other higher order motor areas like the supplementary motor area and cerebellum, directly addressing the central research question of mapping the brain areas differentially activated by hand and foot movements. In contrast, the PCA components reflect broader variance patterns—some related to task lateralization or visual cues—but do not distinctly isolate motor regions. Thus, while Multi Variate Pattern Analysis can reveal complex distributed patterns even in the absence of a task, GLM provides clearer and anatomically interpretable results and should be preferred for this task.

We should apply GLM to test specific hypotheses tested by a suitably designed experiment. It provides more interpretable results and is ideal for hypothesis-driven data analysis. PCA has the advantage of being applicable even in the absence of a task and is useful to decode distributed patterns of activation across voxels rather than focusing on activation in each voxel. It is ideal for data-driven exploratory analysis.

# 3 Appendix

### 3.0.1 Team contributions

| Name | Tasks / Contributions |
|---|---|
| Angana Mondal | Coregistration, PCA |
| Arnault Stähli | Functional Preprocessing |
| Heliya Shakeri | Structural Preprocessing |
| Khushi Singh | Gaussian Smoothing, PCA, Slides |
| Pamela van den Enden | General Linear Model |

**Table 3.1**
Summary of contributions.

### 3.0.2 Brain mask

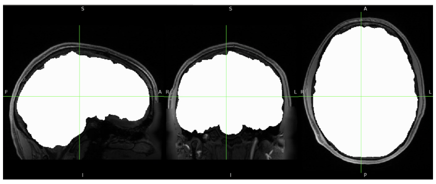


**Figure 3.1**
Binary brain mask generated by BET.

### 3.0.3 Coregistration (bonus)

Coregistration refers to aligning functional data (EPI sapce) with high-resolution anatomical MRI scans (T1 space) for accurate localization of functional activity in the brain. We used the `fslroi` function on the middle volume of the fMRI sequence as a reference to obtain the co-registration matrix. We then used `epi_reg` (boundary-based registration method) to perform the coregistration which gave us a good overall fit for core brain structures, despite some misalignment at the edges which can be attributed to the difference in the resolution of the images and noise from previous preprocessing steps like segmentation.

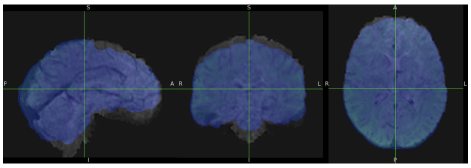


**Figure 3.2**
Fucntional to Anatomical coregistration

### 3.0.4 PCA RESULTS

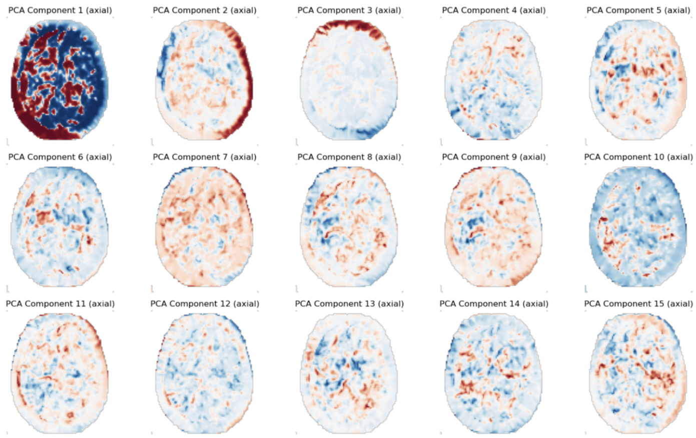



**FIGURE 3.3**
Axial view of the first 15 PCA components.

# Bibliography

Yeo, BT Thomas et al. (2011). 'The organization of the human cerebral cortex estimated by intrinsic functional connectivity'. In: *Journal of Neurophysiology*.

Buckner, Randy L et al. (2011). 'The organization of the human cerebellum estimated by intrinsic functional connectivity'. In: *Journal of Neurophysiology* 106.5, pp. 2322–2345.